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Lung Cancer Screening Disparities in Asian American Subgroups in a Large Integrated Health System

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Abstract

Background: Lung cancer is the leading cause of cancer-related deaths worldwide in Asian Americans (AsA), yet AsA lung cancer screening (LCS) rates are unknown. We examined LCS rates in AsA within Kaiser Permanente Northern California (KPNC), a large integrated healthcare system where LCS is a member benefit. The California LCS rate is 0.7%.

Methods: This cohort study analyzed KPNC 2015-2022 electronic health records. Lung cancer screening rates were compared among AsA subgroups, controlling for sociodemographics, considering both more restrictive 2013 (n = 2,273) and more inclusive 2021 (n = 5,823) United States Preventive Services Task Force (USPSTF) LCS guidelines, which differ by age range and years post-smoking cessation.

Results: Overall KPNC LCS rates for eligible AsA patients were 4.3% and 2.7% using USPSTF 2013 and 2021 guidelines, respectively. Lung cancer screening rates varied by AsA subgroup. Under 2021 guidelines, Chinese (4.0%) were screened more than Korean (3.57%), Southeast Asian (3.52%), Japanese (3.19%), Asian (Other) (2.28%), Pacific Islander (1.91%), and Filipino (1.55%). Under 2013 guidelines, Southeast Asian (6.54%) were screened more than Chinese (6.51%), Japanese (5.36%), Asian (Other) (3.95%), and Filipino (1.93%).

Discussion: This is the first study to demonstrate significant heterogeneity in LCS rates for disaggregated AsA subgroups. Kaiser Permanente Northern California LCS rates were 4× California rates. When payment alone is not a care barrier, systemic and culturally sensitive interventions are necessary to increase overall LCS screening rates and address population-specific disparities.

Keywords

Asian Health Disparities, Lung Cancer Screening, Integrated Healthcare System

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Key Takeaways

- Asian American lung cancer screening rates vary by ethnic subgroup at a large integrated health care system where appropriate lung cancer screening is an included benefit.
- At a large integrated health care system, guideline-directed screening rates remain comparable to the low national level but 4x the state rate.
- Using the 2021 US Preventive Services Task Force guidelines, Chinese individuals had the highest screening rates amongst Asian American subgroups, while Filipino individuals were the lowest screened.
- System-level interventions and culturally sensitive interventions may reduce these disparities, bringing lung cancer screening rates to levels comparable to those of breast and colon cancer screening.

Introduction

Lung cancer is the leading cause of cancer-related deaths worldwide, yet screening rates remain the lowest of all major cancers. In[1] 2020 alone, 1.8 million deaths were attributed to lung cancer—exceeding breast, colon, and prostate cancer mortalities combined. Although recent 5-year lung cancer survival rates are only 23%, substantial improvement has been made from higher mortality rates in the 1960s and 1970s.[2] Early lung cancer detection is directly related to better outcomes and longevity.[3] Currently, lung cancer screening (LCS) is conducted with annual low-dose computed tomography (LDCT), which is low-risk and non-invasive.[3] Despite contributing the most cancer-related deaths, lung cancer remains among the lowest screened (4.5%) compared to breast (75.9%), colon (71.8%), and prostate (37.1%) cancers.[4-7] Nationally, approximately 4.5% of eligible individuals are screened for lung cancer, with some states, such as California, reporting screening rates of 0.7%.[4] These lower rates are attributed to lack of provider knowledge, patient screening access barriers, and other sociodemographic factors.[8,9]

Asian Americans (AsA) are the fastest growing minority group in the United States.[10] However, AsA have disproportionately lower screening rates across multiple cancers, with non-Hispanic white individuals nearly twice as likely to undergo prostate cancer screening compared to their AsA counterparts.[2] Similarly, only 70% of AsA obtained cervical cancer screening (recent Papanicolaou smear) compared to 81% in non-Hispanic white females.[11] Although AsA subgroups have known heterogeneity in biological predisposition and health outcomes, AsA disease prevalence and health risk statistics are often reported as an aggregate population.[12] This reporting practice obscures health disparities within AsA subgroups and masks the need for targeted interventions influenced by cultural differences, language barriers, and health literacy. Aggregate race-specific outcome reporting and model minority stereotypes also conceal health risks of AsA individuals from disadvantaged financial backgrounds with greater lung cancer incidence and mortality risk.[13,14]

Worldwide, 80-90% of lung cancers are due to smoking, with the remainder attributed to genetic mutations (such as eGFR), air quality, and other causes.[15] Current guidelines only target individuals with prior smoking history. In 2013, the United States Preventive Services Task Force (USPSTF) recommended LCS for individuals 55 to 80 years old, with a 30 pack-year smoking history, who were current smokers or had quit within 15 years. Based on the 2020 NELSON trial, in 2021, the USPSTF increased screening eligibility by expanding eligible patient age range to 50-80 years old and reducing the required smoking burden to 20 pack-years.[16]

As with other cancer screening trends within homogenized populations, we hypothesized LCS patterns may vary within disaggregated racial groups by both 2013 and 2021 USPSTF screening guidelines. Our Kaiser Permanente Northern California (KPNC) health system was uniquely poised to explore this issue in the AsA community. First, KPNC is an integrated health system with multiple clinics focused on primary prevention. Second, in the San Francisco Bay Area, almost 30% of patients identify as Asian. Third, most KPNC AsA patients self-identify their AsA racial/ethnic patient demographic information, and KPNC has longitudinal LCS data on all at-risk patients. We explored LCS rates in disaggregated at-risk AsA, controlling for age, smoking history, clinical, and sociodemographic factors, in comparison to the state of California's screening uptake rates.

Methods

Study Design and Population

We conducted a retrospective longitudinal cohort study exploring LCS rates in a large integrated health care system, Kaiser Permanente Northern California (KPNC), with screening eligible AsA patients enrolled from January 1, 2015 to December 31, 2022, by reviewing electronic health record data (Epic™, Verona, WI). This was approved by the KPNC institutional review board (protocol 2250470-1).

Kaiser Permanente Northern California services a population of 4.6 million members in Northern California with more than 40% of the insured population, with 600 primary and specialty clinics combined.[17] Kaiser

Permanente Northern California utilizes an integrated health care model where enrolled patients automatically have insurance coverage, improving efficiency and patient-provider outcomes.[8]

There were 245 905 AsA patients who were smokers or former smokers in the original data set. After removing those without complete smoking information, 19.7% (n = 48 442) remained. We included AsA patients meeting 2013 USPSTF guideline screening eligibility (55 to 80 years old, 30 pack-years history, current smoker, or 15 years since quitting) and the 2021 USPSTF guideline screening eligibility (50 to 80 years old, 20 pack-years history, current smoker, or 15 years since quitting). Under the 2013 guidelines, 2283 (4.7%) of AsA patients were eligible, and using the expanded 2021 guidelines, which included the previous 2013 cohort in addition to all newly eligible patients, 5823 (12.0%) AsA individuals remained.

Asian Disaggregation and Covariates. At KPNC enrollment, patients self-identify health behaviors and demographics, including their racial/ethnic subgroup. We included patients identifying as Asian (Chinese, Filipino, Pacific Islander, Japanese, Korean, Southeast Asian (mainly Vietnamese, 93% of category), and Asians without subgroup selection). We created an "Asian (Other)" category to include Asians without subgroup selection and those with subgroup sample sizes less than ten (including Asian Indians (n = 44)). Frequencies are reported in Table 1.

We confirmed Asian subgroup self-identification on 20% of the sample, using medical record number verification with patient self-reported Asian origin ancestry.

Covariates included age by years (50-60, 61-70, and 71-80), sex assigned at birth (male, female), smoking status (current, former), and language preference (English, not English). As the number of AsA varies by California county, we included 21 patient clinic locations, collapsed into 5 geographic regions: Region 1 (all clinics in Roseville, South Sacramento, Sacramento), Region 2 (Santa Clara, San Jose, Redwood City), Region 3 (Central Valley, Fresno), Region 4 (Diablo, Napa, Solano), and Region 5 (East Bay, San Francisco, Marin, Solano). Socioeconomic status was proxied by the Neighborhood Deprivation Index (NDI quartiles, 1 = least deprived, 4 = most deprived), which classifies poverty level according to individual zip code and 13 socioeconomic variables from the American Community Survey. Pack-years and years smoked were utilized only to determine patient eligibility for the 2013 or 2021 cohorts and were not included in the regression analysis.

Table 1. Baseline Characteristics, Asian Subgroup

Characteristic	Chinese (n = 1363)	Filipino (n = 1028)	Pacific islander (n = 366)	Korean (n = 140)	Japanese (n = 251)	South Asian (n = 312)	Asian (other) (n = 2363)	Total (n = 5823)
Age group (years)								
50-60	681	406	221	67	67	186	1417	3044
61-70	516	434	122	48	114	108	798	2140
71-80	166	188	23	25	70	18	148	639
Sex								
Male	1261	811	262	107	154	306	2012	4913
Female	102	217	104	33	97	6	351	910
Smoking status								
Current	1305	876	329	134	203	303	2200	5350
Former	58	152	37	6	48	9	163	473
Language preference								
English	306	979	341	22	228	38	2187	4101
Non-English	1057	49	25	118	23	274	176	1722
Region								
Region 1	154	115	83	27	62	65	396	902
Region 2	170	147	69	39	54	130	501	1110
Region 3	35	68	44	5	18	11	199	380
Region 4	78	243	71	15	28	10	380	825
Region 5	926	455	99	54	89	96	887	2606
Neighborhood Deprivation Index (NDI)								
Quartile 1	216	68	37	40	70	19	354	821
Quartile 2	435	323	96	54	88	66	722	1795
Quartile 3	421	373	110	33	54	102	721	1816
Quartile 4	291	264	123	13	39	125	566	1391

Lung Cancer Screening Completion and Follow-Up Period. Screening was considered completed if at least 1 lung screening test (LDCT lung) was ordered by a KPNC provider (physician or advanced practice practitioner) during the study period. We excluded those with lung cancer or with less than 6 months of continuous KPNC membership after self-reporting smoking status. Radiologic tests including standard lung CT, PET, or fMRI imaging were considered diagnostic (symptom directed or follow-up testing) and were excluded as primary LCS modalities.

Follow-up time was defined as the period from a participant's initial eligibility date for lung cancer screening, as determined by the 2021 USPSTF guidelines, until the first occurrence of any of the specific events: completion of initial LDCT, censoring on the date of disenrollment from the KPNC health plan, death, or study period end-date.

Analytic Methods

We conducted univariate (N and frequencies) analysis of eligible AsA subgroup patients for both 2013 and 2021 guidelines, including demographics (age, sex, language preference, socioeconomic status (NDI)), and health behaviors (smoking)). Cox proportional hazard regression estimated association between completed LCS and participants' sociodemographic characteristics, controlling for age (71-80 years as reference), AsA subgroup (Chinese), NDI (1, least deprived), language (English), and patient region (Region 5) for the 2021 cohort. We report unadjusted and adjusted hazard ratios with respective 95% CI. Analysis was conducted with v9.4, SAS Institute Inc, Cary, NC), with statistical significance at $p < 0.05$.

Results

Eligibility by Screening Guidelines

Using the 2013 USPSTF LCS guidelines, a total of 2273 AsA KPNC patients were eligible for LCS by LDCT (Figure 1). Patients were 84.6% male, aged 63 years (mean 62.8 ± 5.3), with 68.7% preferring English. Eligible AsA subgroups included Asian (Other) (n = 861, 37.9%), Chinese (584, 25.7%), Filipino (414, 18.2%), Pacific Islander (133,

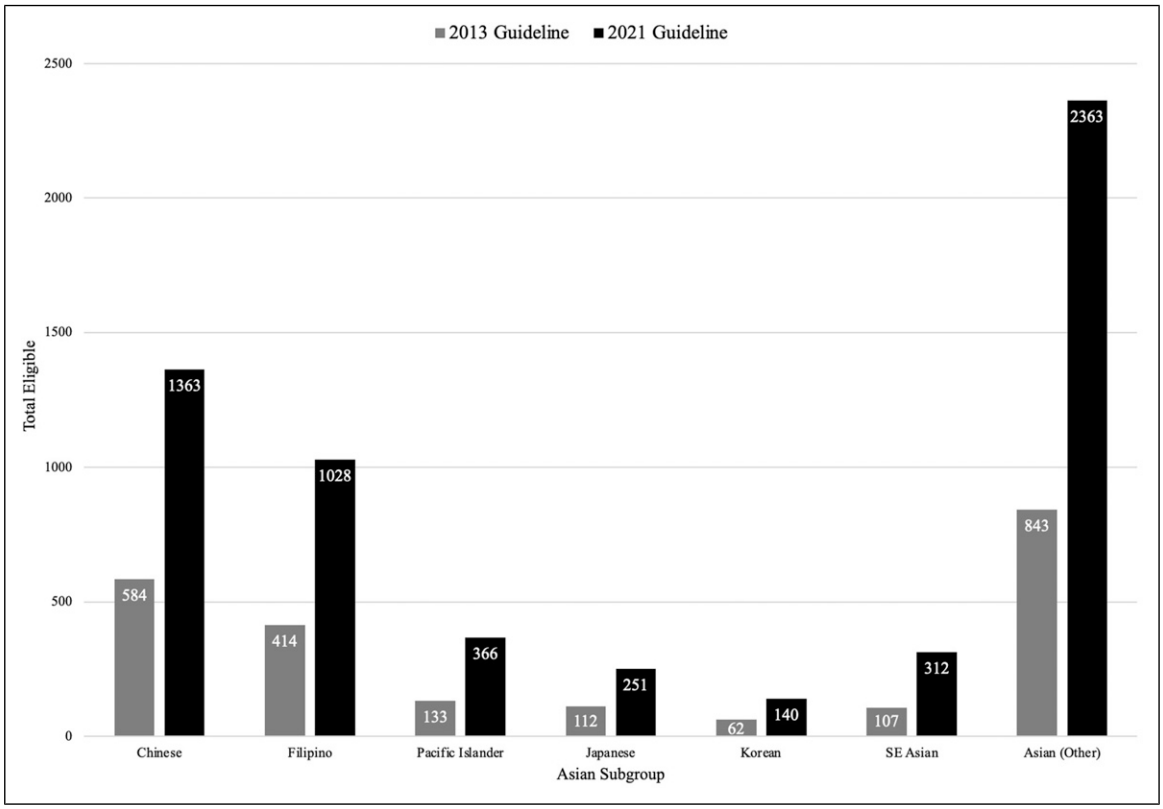


Figure 1. Lung Cancer Screening Eligibility by USPSTF Guidelines per Asian Subgroup. Abbreviations: USPSTF, United States Preventive Services Task Force; SE, Southeast. Lung Cancer screening eligibility separated by Asian subgroup per the USPSTF guideline in 2013 in gray (55-80 years old; current or former smoker with 30 pack-year smoking history, or 15 years since quitting) versus 2021 guideline in black (50 -80 years; 20 pack-year smoking history, or 15 years since quitting).

5.9%), Southeast Asian (107, 4.7%), Japanese (112, 4.9%), and Korean (62, 2.7%).

The USPSTF 2021 LCS guidelines increase screening-eligible patients by increasing the eligibility age range by 5 years and reducing pack-year history by 10 years. Using 2021 USPSTF guidelines, 5823 AsA patients were deemed eligible for LCS, a 156% increase from 2013 guidelines. Eligible patients were 83.6% male, aged 61 years (mean 61.1 ± 7.0), and 70.2% preferring English. Eligible AsA subgroups included increased by over 100 each, with Asian (Other) (n = 2363, 40.6%), Chinese (1363, 23.4%), Filipino (1028, 17.6%), Pacific Islander (366, 6.3%), Southeast Asian (312, 5.4%), Japanese (251, 4.3%), and Korean (140, 2.4%).

Lung Cancer Screening Rates by Disaggregated Asian Subgroups by Screening Guidelines

Using the 2013 USPSTF guidelines, the KPNC Asian aggregate LCS rate was 4.31%. Southeast Asian patients had the highest rate at 6.54%, above Chinese at 6.51%, Japanese at 5.36%, Asian (Other) at 3.95%, and Filipino at 1.93% (Figure 2). Pacific Islander and Korean patients were not reported for 2013 guidelines due to sample sizes.

Using the 2021 USPSTF guidelines, the KPNC Asian aggregate LCS rate was 2.66%. Chinese patients had the highest LCS rate at 3.96%, above Korean at 3.57%, Southeast Asian at 3.52%, Japanese at 3.19%, Asian (Other) at 2.28%, Pacific Islander at 1.91%, and Filipino at 1.55% (Figure 3).

Predictors of LCS by Sociodemographic Factors. Using Cox proportional hazard regression analysis, patients aged 61-70 years had higher screening uptake compared to older patients (71-80 years) in both univariate (HR: 2.10, 95% CI: 1.12-3.96, $P = 0.021$) and multivariable models (HR: 2.08, 95% CI: 1.10-3.94, $p = 0.024$). No differences were observed for those aged 50-60 years. Filipino individuals were much less likely to be screened than Chinese individuals (reference), in both univariate (HR: 0.35, 95% CI: 0.20-0.62, $p < 0.001$) and multivariable models (HR: 0.37, 95% CI: 0.19-0.71, $p = 0.003$). The Asian (Other) subgroup also had reduced screening uptake compared to the Chinese subgroup in both univariate (HR: 0.59, 95% CI: 0.41-0.86, $p = 0.006$) and multivariable models (HR: 0.58, 95% CI: 0.35-0.96, $p = 0.033$). No other AsA subgroups differed significantly from Chinese.

Lung cancer screening uptake was not significantly associated with NDI quartiles in the adjusted model. Compared to English-speaking patients, Non-English speakers showed a trend toward higher screening uptake in the univariate model (HR: 1.68, 95% CI: 1.22-2.32, $p = 0.002$),

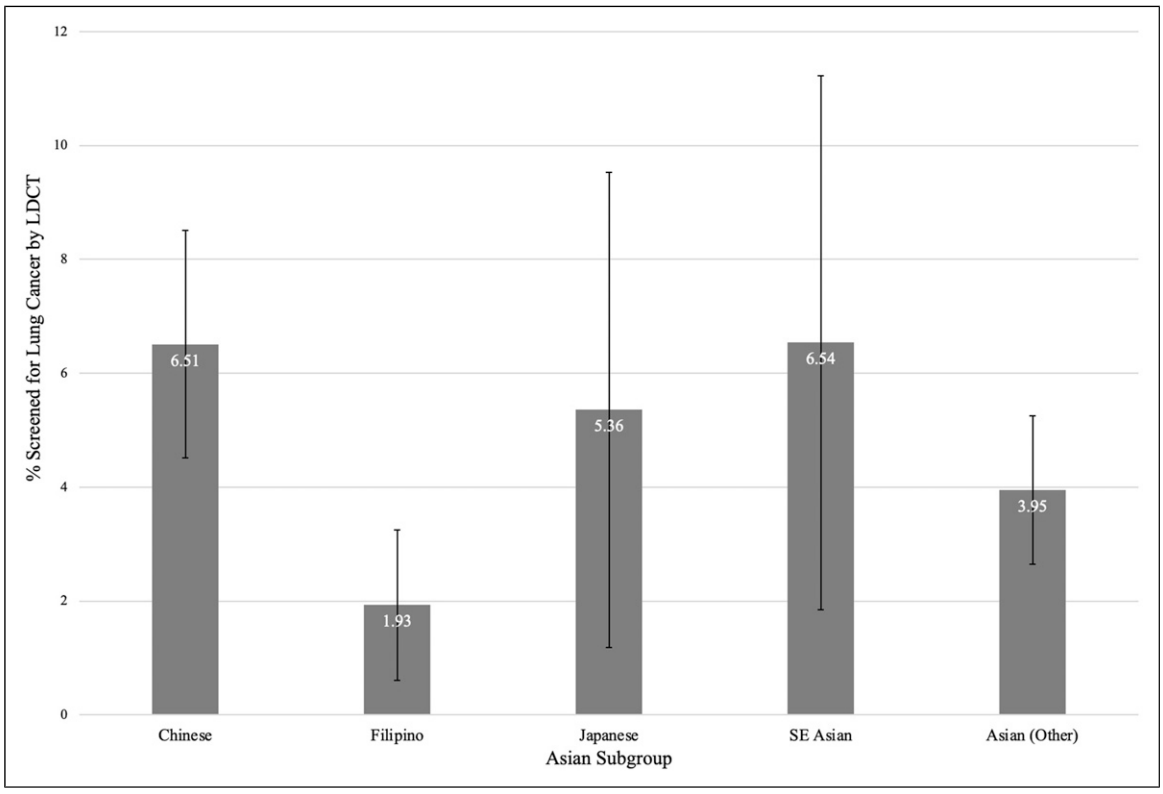


Figure 2. Lung Cancer Screening Disaggregated by Asian Under 2013 USPSTF Guidelines. Abbreviations: USPSTF, United States Preventive Services Task Force; LDCT, Low-Dose Computed Tomography; SE, Southeast. Screening rates by comparative bar chart, per USPSTF 2013 screening guidelines, stratified by Asian ethnic subgroups.

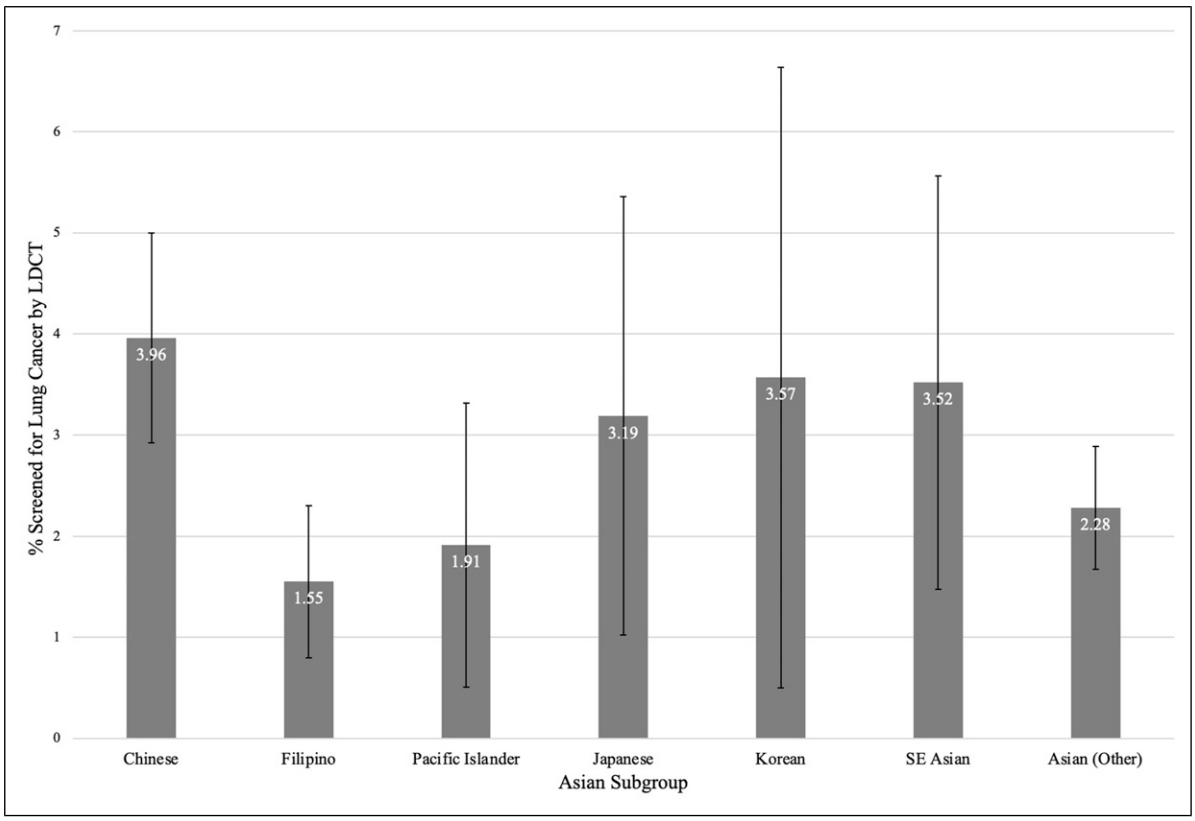


Figure 3. Lung Cancer Screening Disaggregated by Asian Subgroup Under 2021 USPSTF Guidelines. Abbreviations: USPSTF, United States Preventive Services Task Force; LDCT, Low-Dose Computed Tomography; SE, Southeast. Screening rates by comparative bar chart, per USPSTF 2021 screening guidelines, stratified by Asian ethnic subgroups.

but this association was non-significant in the multivariable model (HR: 1.16, 95% CI: 0.72-1.86, $p = 0.535$). Only Region 3 patients had a significantly higher LCS rate compared to Region 5, in both univariate (HR: 2.28, 95% CI: 1.38-3.76, $p = 0.001$) and multivariable models (HR: 2.92, 95% CI: 1.72-4.97, $p < 0.001$). The complete regression model results are reported in Table 2.

Discussion

We found overall low rates of lung cancer screening for eligible AsA within our health system, at 2.7%. While low, these rates nearly quadruple the California LCS average of 0.7%. Using current 2021 USPSTF LCS guidelines, Chinese individuals had the highest screening rates at 4.0%, more than twice the rate of Filipino individuals at 1.6%. Older patients (71-80 years) had half the screening rates of younger patients. Current or prior smoking did not affect LCS. Geographic region influenced LCS rate as patients from Region 3, the most economically disadvantaged, were more likely to undergo LCS. This is likely from the robust registered nurse (RN) navigator program only in this region.

Despite the number of LCS-eligible individuals increasing in 2021, LCS rates did not significantly improve in our cohort. Between more inclusive standards and newly established navigator programs, a greater quantity of patients were screened, but in a lower proportion compared to the expanded eligible population. Consequently, more patients received LCS but at a lower rate than in 2013, likely from lack of provider awareness on the recent guideline changes. Educational efforts on the recent USPSTF guidelines are ongoing.

Our findings highlight the benefits of an integrated health care model for LCS. All KPNC AsA subgroups had significantly higher LCS rates than the state-wide rate.[4] Importantly, an individual socioeconomic status was not associated with lung cancer screening—a key KPNC patient benefit. Additionally, Region 3, where 20% of residents live under the poverty line,[18] ranked best in LCS uptake among eligible Asian patients, with 2-3 fold higher LCS rates than all other regions—a result of the nurse navigator program. This study corroborates others on the effect of centralized screening programs in improving patient outcomes.[18]

Our results also support the significant heterogeneity within LCS eligibility and uptake rates by AsA subgroups, with the highest screening observed among Chinese individuals and the lowest in Filipino individuals. While Filipino individuals are generally among the highest earning subgroups, medical mistrust, fatalistic misconceptions, and pervasive stigmas about cancer screening and diagnosis may reduce LCS rates.[10,19] Conversely, Chinese individuals may

Table 2. Univariate and Multivariate Cox Proportional Hazards Regression Analyses of Factors Associated With Lung Cancer Screening Uptake Among Eligible Adults

	Univariate analysis		Multivariable analysis	
Characteristic	HR (95% CI)	*P*-Value	HR (95% CI)	*P*-Value
Age group (years)				
50-60	1.38 (0.73-2.61)	0.32	1.38 (0.72-2.64)	0.33
61-70	2.10 (1.12-3.96)	0.02	2.08 (1.1-3.94)	0.02
71-80	1.0 [reference]		1.0 [reference]	
Race/Ethnicity				
Asian (other)	0.59 (0.41-0.86)	0.006	0.58 (0.35-0.96)	0.03
Pacific Islander	0.50 (0.23-1.09)	0.08	0.47 (0.2-1.12)	0.09
Filipino	0.35 (0.2-0.62)	0.0003	0.37 (0.19-0.71)	0.003
Japanese	0.69 (0.33-1.45)	0.32	0.65 (0.29-1.47)	0.30
Korean	0.88 (0.35-2.2)	0.78	0.80 (0.32-2.0)	0.64
SE Asian	0.99 (0.52-1.9)	0.98	0.90 (0.46-1.77)	0.76
Chinese	1.0 [reference]		1.0 [reference]	
Neighborhood Deprivation Index (NDI)				
Quartile 1	1.0 [reference]		1.0 [reference]	
Quartile 2	0.68 (0.42-1.1)	0.12	0.69 (0.42-1.12)	0.14
Quartile 3	0.69 (0.43-1.12)	0.13	0.70 (0.43-1.16)	0.17
Quartile 4	0.83 (0.51-1.35)	0.44	0.77 (0.46-1.3)	0.32
Language preference				
English	1.0 [reference]		1.0 [reference]	
Non-English	1.68 (1.22-2.32)	0.002	1.16 (0.72-1.87)	0.54
Region				
Region 1	1.09 (0.67-1.75)	0.74	1.19 (0.73-1.95)	0.49
Region 2	1.23 (0.8-1.88)	0.35	1.309 (0.84-2.05)	0.24
Region 3	2.28 (1.38-3.76)	0.001	2.92 (1.72-4.97)	<.0001
Region 4	0.84 (0.49-1.43)	0.51	1.09 (0.63-1.91)	0.74
Region 5	1.0 [reference]		10 [reference]	

experience higher screening rates as preventive medical practices and trust in health care are more culturally supported. Specifically, Chinese individuals had the highest screening rate under the 2021 guidelines, while Southeast Asians had the highest screening rate under the 2013 guidelines. We conjectured that Chinese individuals have a greater propensity to smoke but have less persistent smoking habits compared to Southeast Asians, allowing a greater proportion of Chinese individuals to be included under the more lenient guidelines. Patients aged 61-70 were more likely to receive lung cancer screening than those aged 71-80 likely from no competing comorbidities that may preclude them from pursuing a discretionary preventive measure like cancer screening.

This study has several strengths, including longitudinal data, a diverse patient population, and an EHR with disaggregated racial and ethnic data. It was recently verified that the KPNC registry is more diversely representative and ideal for studying real-world lung cancer patients compared to National Cancer Institute designated cancer centers.[20] Study limitations may reduce generalizability. While patients had LCS as a covered benefit, those with high out-of-pocket expenses in other health systems or insurances may limit screening utilization. Patients self-reported smoking history, with potential under reporting from social desirability bias. Additionally, over 80% of AsA patients who were smokers or former smokers were excluded from our analysis due to missing pack-years smoked and years-since-quitting data, potentially introducing selection bias. Stratification by sex was not possible due to sample size constraints with more male smokers.

In this study, LCS rates varied significantly across eligible AsA subgroups, with Chinese individuals recording the highest rate. This can be attributed to intrinsic cultural factors favoring preventive medical approaches, KPNC's integrated health care system, and robust nurse navigator awareness programs at some KPNC clinics. The AsA aggregate screening rate under the most recent guidelines significantly exceeded the California average, and this model could help inform future efforts to improve LCS rates state-wide. Overall, our study supports the significant progress that has been made in improving preventive approaches, especially in reaching AsA subpopulations.

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Ethical Considerations

This study was approved by the KPNC institutional review board (protocol 2250470-1]).

Consent to Participate

All electronic health record data was de-identified.

Author Contributions

Conceptualization: Jeffrey B. Velotta, Lester Andrew V. Uy, Armaan Jamal, Malathi Srinivasan, Nitya Rajeshuni, Gloria S. Kim, Robert J. Huang, and Latha Palaniappan. Data curation: Jeffrey B. Velotta and Sheng-Fang Jiang. Formal analysis: Sheng-Fang Jiang. Funding acquisition: Malathi Srinivasan, Gloria S. Kim, and Latha Palaniappan. Investigation: Jeffrey B. Velotta and Carmen Javier. Methodology: Sheng-Fang Jiang, Seth J. Tivakaran, Julia L. Raghu, Jeff CK. Leung, King Sum Tong, Nicholas L. Panyanouvong, Carmen Javier, Jeffrey B. Velotta, and Malathi Srinivasan. Project administration: Jeffrey B. Velotta, Malathi Srinivasan, Nitya Rajeshuni, and Gloria S. Kim. Software: Sheng-Fang Jiang. Supervision: Jeffrey B. Velotta, Malathi Srinivasan, Nitya Rajeshuni, Gloria S. Kim, Robert J. Huang, and Latha Palaniappan. Validation: Sheng--Fang Jiang. Visualization: Julia L. Raghu. Writing—original draft: Seth J. Tivakaran, Julia L. Raghu, Jeff CK. Leung, King Sum Tong, Nicholas L. Panyanouvong, Carmen Javier, and Sheng-Fang Jiang. Writing—review and editing: Seth J. Tivakaran, Julia L. Raghu, Jeff CK. Leung, King Sum Tong, Nicholas L. Panyanouvong, Carmen Javier, Sheng-Fang Jiang, Malathi Srinivasan, Jeffrey B. Velotta, and Gloria S. Kim.

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Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement

The data reported was proprietary electronic health record data from the Kaiser Permanente Division of Research database.

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